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# Optimizing Scientific Manuscript Preparation for Scopus-Indexed Publication: A Comprehensive, Critical Review of Best Practices, Pitfalls, and Research-Validated Strategies

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**Systematic Review**

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**REVIEW ARTICLE**

# OPTIMIZING SCIENTIFIC MANUSCRIPT PREPARATION FOR SCOPUS-INDEXED PUBLICATION: A COMPREHENSIVE, CRITICAL REVIEW OF BEST PRACTICES, PITFALLS, AND RESEARCH-VALIDATED STRATEGIES

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***Running head :*** *Optimizing Scientific Writing for Scopus-Indexed Manuscripts*

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## ABSTRACT

The growing demand for publication in Scopus-indexed journals has reshaped the priorities of researchers, particularly in the medical sciences, where academic advancement and institutional reputation are closely tied to indexed output. Yet, despite extensive guidance on scientific writing, no review has critically synthesized the diverse methodological, ethical, and practical challenges that shape manuscript preparation in the contemporary era of artificial intelligence. This systematic review follows PRISMA 2020 recommendations to evaluate current evidence on best practices, common pitfalls, and emerging strategies that influence successful publication. Searches across major databases and registers identified 1,462 records, of which 39 studies met eligibility criteria. Included literature addressed core domains of manuscript structure, journal selection, research integrity, AI-assisted writing, reviewer expectations, and the evolving landscape of predatory publishing. Findings reveal that successful submissions stem from a convergence of conceptual clarity, methodological rigor, ethical transparency, and strategic alignment with journal scope. Contrary to common assumptions, Scopus acceptance is influenced less by article processing charges or co-authorship prestige and more by adherence to guidelines, clarity of argumentation, and demonstrable novelty. AI tools offer meaningful improvements in organization and linguistic refinement but carry ethical constraints, particularly regarding undisclosed use, fabricated citations, and authorship attribution. Across studies, consistent themes emerged: the importance of humanizing narrative tone, maintaining transparency in AI involvement, and preparing figures, tables, and references early in the drafting process. This review provides an integrated, evidence-informed blueprint for authors seeking to optimize manuscript preparation and navigate the increasingly complex pathway toward Scopus-indexed publication.


**Keywords:** *Scientific writing, Scopus-indexed journals, Manuscript preparation, Artificial intelligence ethics, Publication best practices*

## INTRODUCTION

Publishing in Scopus-indexed journals has become a central ambition for medical researchers, driven by strong academic incentives and institutional expectations that position indexed publications as markers of scientific credibility, visibility, and career progression. This emphasis reflects a longstanding belief that indexed journals maintain higher standards of methodological rigor, clarity of argumentation, and editorial oversight than non-indexed outlets, a belief reinforced by decades of instructional literature emphasizing the importance of coherent structure, logical flow, and strong scientific storytelling in manuscript preparation.[1] The need for polished academic writing is repeatedly underscored, as researchers recognize that publication in respected platforms advances both personal trajectories and broader disciplinary knowledge.[2] Yet, despite widespread awareness of these motivations, uncertainty persists about what truly differentiates successful submissions, prompting renewed calls for refined guidance on writing practices and publication strategy.[3]

Recent scholarship has highlighted the complexity of academic publishing and the subtle barriers that discourage or delay submissions, including misconceptions about peer review, anxiety regarding manuscript rejection, and challenges associated with presenting arguments with

sufficient logical coherence.[4] These apprehensions are further intensified by discussions surrounding predatory journals and Beall's List, a controversial but influential resource that attempted to categorize journals based on deceptive editorial practices.[5] Although originally designed to warn scholars, debates regarding the validity, accuracy, and boundaries of the list continue, with some analyses noting that journals indexed in reputable databases may still exhibit questionable characteristics, especially during transitional periods of quality control.[6] Such inconsistencies amplify concerns about where to publish and how to assess journal legitimacy, reinforcing the perception that publishing in Scopus-indexed venues requires exceptional precision and awareness of evolving editorial norms.[7]

Authors frequently question whether financial factors, including article processing charges, influence acceptance and whether collaboration with established researchers offers a meaningful advantage. Empirical discussions, however, demonstrate that high-quality writing, strong methodological grounding, and adherence to journal guidelines exert greater influence on editorial decisions than financial or relational considerations.[8] Improving clarity, coherence, and structural integrity remains central to favorable reviewer impressions, a theme consistently reiterated in methodological guidance across biomedical fields.[9] These recommendations align with contemporary proposals for stepwise approaches to scientific writing that emphasize planning, conceptual refinement, and systematic drafting.[10]

The rapid rise of artificial intelligence has introduced new opportunities and challenges for manuscript development. Emerging consensus statements stress the need for transparency in reporting AI use, emphasizing that such tools can support writing but cannot replace human reasoning, critical interpretation, or authorship responsibility.[11] Ethical concerns surrounding AI misuse, including plagiarism, hallucinated citations, and fabricated data, have been documented extensively, prompting calls for caution and responsible integration.[12] Experimental evidence suggests that even trained reviewers struggle to distinguish AI-generated content from human writing, raising broader questions about authenticity and accountability in scholarly communication.[13] These concerns intersect with long-standing principles of scientific writing, which emphasize clarity, coherence, methodological soundness, and integrity.[14] Responsible engagement with AI tools therefore requires adherence to established ethical frameworks, transparent declaration of assistance, and careful human oversight.[15]

The broader publication landscape further shapes researcher behavior, as comprehensive guides on publishing continue to highlight the importance of understanding the scope and expectations of target journals.[16] Step-by-step frameworks for writing scientific manuscripts underscore the value of conceptual planning, reader-oriented structure, and disciplined revision.[17] Educational resources reinforce these principles, emphasizing clarity, logical development, and alignment between research questions, methods, and conclusions.[18] These expectations resonate across decades of scholarly discourse, consistently emphasizing reader comprehension and narrative coherence.[19] Even foundational guides aimed at novice researchers stress the importance of organizing ideas, building arguments incrementally, and approaching writing as an iterative scientific process.[20] Clarity and precision are universal requirements, anchoring the expectations of authors and editors alike.[21]

At the editorial level, publication standards continue to evolve. Professional guidelines outline expectations for research integrity, transparency, and ethical authorship, reinforcing norms that shape manuscript evaluation and editorial decision-making.[22] Studies examining barriers to manuscript completion identify limited time, insufficient mentorship, and unfamiliarity with writing conventions as major obstacles—challenges that frequently contribute to delays or lower-quality

submissions.[23] Concerns about AI-generated manuscripts further complicate the landscape, as reviewers demonstrate variable accuracy in identifying such content, particularly when writing is technically polished but lacks conceptual depth.[24] Simultaneously, thematic analyses of publisher guidelines reveal substantial variability in permitted AI uses, indicating an active period of adaptation as journals refine their policies.[25] Technical proposals for AI-assisted authoring frameworks highlight the need for infrastructure that supports transparency, traceability, and responsible use.[26]

Ethical reflections continue to warn against unacknowledged AI contributions and emphasize the importance of maintaining human judgment at every stage of scientific communication.[27] Editorial guidance stresses that structure, coherence, and readability remain central determinants of acceptance and cannot be replaced by automation.[28] Expert commentaries add nuance by outlining common reasons for manuscript rejection and suggesting strategies to enhance clarity, coherence, and scientific rigor.[29] Broader discussions of AI-related authorship challenges reiterate that AI cannot serve as an author and must remain a tool subordinate to human decision-making.[30] Leading commentaries further outline priorities for AI-related research, underscoring the need for transparency, safety, and responsible governance.[31] These evolving perspectives influence how authors must navigate writing tools, plagiarism screening, and authenticity safeguards—particularly as AI use in academic environments expands.[32]

Methodological reflections emphasize that writing quality, ethical awareness, peer review responsiveness, and clarity of communication remain foundational to maintaining trust in scientific publishing.[33] Concerns about predatory journals continue to surface, with analyses documenting the systemic vulnerabilities faced by researchers—especially early-career authors navigating complex publication environments.[34] Expert perspectives consistently reiterate core qualities of strong manuscripts, including structured argumentation, coherent data presentation, and reader-oriented clarity.[35] The growing conversation about AI, authorship, and academic ethics reflects an ongoing cultural shift in how scientific writing is conceptualized and evaluated.[36] Broader ethical dialogues highlight the intersection between AI, peer review, and authorship integrity, reinforcing the need for robust normative frameworks.[37] Practical commentaries outline how AI should be used cautiously, transparently, and under human supervision to avoid compromising scientific integrity.[38] Foundational guides for novice researchers continue to provide essential strategies for writing, revising, and navigating the publication process, further contributing to an integrated understanding of best practices.[39]

Although publication timelines vary widely across disciplines and journal quartiles, most Scopus-indexed journals follow a broadly predictable trajectory from submission to acceptance. The initial editorial screening, during which manuscripts are checked for scope alignment, structural adequacy, and ethical compliance, typically requires between one and three weeks. The subsequent peer-review period is the most variable stage, often ranging from six to twelve weeks depending on reviewer availability and the complexity of the manuscript. Revised submissions usually undergo a shorter evaluation lasting two to four weeks, though multiple rounds of revision are not uncommon in journals with high competition and methodological expectations. Following acceptance, the production phase—which includes copyediting, typesetting, proofing, and online release—may take an additional four to eight weeks. While fast-track journals occasionally accelerate these intervals, the overall timeline from submission to publication commonly spans three to six months, with certain high-impact outlets extending beyond this window when reviews are delayed or editorial workloads increase.

Despite extensive literature on discrete aspects of scientific writing, no prior review has synthesized these themes into a comprehensive, critical framework tailored to researchers seeking publication in Scopus-indexed medical journals. A systematic review following PRISMA 2020 guidelines is therefore essential to consolidate current evidence, address misconceptions, and identify research-validated strategies for producing manuscripts that meet the methodological, ethical, and editorial standards required for successful publication. This review aims to integrate insights across writing methodology, publication behavior, AI ethics, peer review, and editorial expectations to generate an authoritative blueprint for optimizing scientific manuscript preparation and navigating the contemporary landscape of Scopus-indexed scholarly publishing.

## METHODS

### Study Design

This review was conducted as a systematic review informed by the PRISMA 2020 framework, with the objective of synthesizing current evidence on scientific writing practices, publication strategies, and the evolving ethical landscape surrounding AI-assisted authorship in Scopus-indexed medical journals. The methodological approach was structured to ensure transparency, reproducibility, and rigor, aligning with established expectations for systematic evidence synthesis. Although this review adheres closely to the PRISMA format, it was not registered with PROSPERO due to the absence of a completed protocol at the time of initiation. The scope and complexity of this review required an iterative development process that refined research questions and analytic priorities as the literature base was examined.

### Eligibility Criteria

Studies were eligible for inclusion if they contributed substantive insights into scientific manuscript preparation, journal selection, publication ethics, peer review behavior, or the integration of artificial intelligence into writing workflows. Eligible literature encompassed narrative reviews, commentaries, editorials, consensus statements, methodological guides, experimental studies, survey-based investigations, technical frameworks, and perspectives addressing writing quality, ethical authorship, predatory publishing, or AI governance. Empirical studies evaluating reviewer behavior, author challenges, or detection of AI-generated manuscripts, such as those described by Helgeson and colleagues and Öztürk and collaborators, were also included because they illuminate emerging dimensions of authorship evaluation within contemporary publishing ecosystems. Papers unrelated to scholarly communication, lacking relevance to writing or publication processes, or focused exclusively on clinical outcomes were excluded. These criteria ensured that the final body of literature reflected a coherent and methodologically diverse foundation for thematic synthesis.

### Information Sources

A comprehensive search was conducted across major biomedical and academic databases, including PubMed, PMC, Scopus Preview, CrossRef, and Google Scholar, along with editorial policy registries such as DOAJ and Retraction Watch listings. These sources were selected to capture the wide methodological spectrum represented in writing-focused research and AI-governance literature, reflecting contributions ranging from foundational guidance on manuscript structure to advanced discussions of transparency and ethical disclosure. Figure 1 summarizes the complete flow of identification, screening, eligibility assessment, and inclusion.

## Search Strategy

Search terms encompassed controlled vocabulary and free-text combinations related to scientific writing, manuscript preparation, publication ethics, peer review, predatory journals, Scopus indexing, AI-assisted writing tools, reviewer detection of AI-generated text, and disclosure standards. Filters were not applied for study design, allowing broad representation of the diverse formats characteristic of writing and ethics scholarship. The last search was completed immediately before data extraction to ensure the inclusion of emerging literature reflecting rapid developments in AI policy and editorial expectations.

## Selection Process

All retrieved records were screened in two phases: an initial title and abstract review followed by full-text assessment. Screening was performed by multiple reviewers working independently, with disagreements resolved through deliberation and consensus. Automated tools supported the identification of duplicates and non-academic sources flagged for exclusion. Reports that lacked full-text access or demonstrated ambiguity regarding relevance were discussed collectively to maintain alignment with the predefined eligibility framework. The final selection included thirty-nine studies, with reasons for exclusion documented during full-text review. Table 1 provides an overview of the core characteristics of included studies, reflecting the diversity of methodological approaches, thematic relevance, and risk of bias assessments.

## Data Collection Process

Data extraction was conducted independently by multiple reviewers using a standardized template. Extracted data included study design, research aims, writing or publication topics addressed, ethical considerations, AI-related contributions, perceptions of manuscript difficulty, reviewer behavior, and recommendations for improving scientific communication. Investigators were not contacted for additional information, as all included studies provided sufficient methodological detail for synthesis. No automation tools were used to extract data, in line with the commitment to human oversight emphasized throughout discussions of AI in scholarly writing. Table 2 summarizes methodological rigor, reporting quality, ethical transparency, reproducibility potential, and AI disclosure characteristics across included studies.

## Data Items

Key data elements included outcomes related to manuscript quality, structural coherence, clarity of argumentation, journal selection strategies, ethical authorship, plagiarism screening practices, AI integration, peer review expectations, and determinants of acceptance or rejection in Scopus-indexed journals. Studies describing barriers to writing, such as insufficient mentorship or time constraints, were included because these contextual factors meaningfully influence manuscript readiness. Ethical issues such as predatory publishing, AI hallucinations, deceptive editorial practices, and integrity risks were documented because they shape the writing environment and inform recommended safeguards. When reporting was unclear, assumptions were minimized to preserve fidelity to the original evidence.

## Risk of Bias Assessment

Risk of bias was evaluated qualitatively using criteria appropriate to each study design, given the heterogeneous nature of the included literature. Considerations included transparency of methods, clarity of aims, adequacy of reporting, potential for author or editorial bias, and the extent to which conclusions were supported by evidence. The diversity of formats—from consensus statements and technical frameworks to randomized experiments assessing reviewer accuracy—necessitated a flexible appraisal approach. Table 3 maps thematic contributions across writing technique, publication barriers, ethics, AI integration, peer review, and editorial standards, positioning each study within the conceptual scaffolding of the review.

### Effect Measures

Because the included studies varied considerably in design and purpose, no quantitative effect measures were applied. Instead, the synthesis relied on narrative integration and thematic clustering to identify recurrent patterns and emerging insights relevant to manuscript preparation and publication in Scopus-indexed journals.

### Synthesis Methods

Studies were grouped thematically according to their primary contributions: manuscript structure and writing technique, publication strategy and journal selection, ethical authorship and predatory publishing, peer review dynamics, AI-assisted writing practices, and editorial expectations. Data were synthesized through iterative comparative analysis that examined convergences and divergences across methodological traditions. Preparation of the synthesis included reviewing study narratives, extracting core recommendations, and mapping conceptual relationships among findings using Figures 2 and 3 to support visual integration. Meta-analysis was not feasible due to substantial heterogeneity; instead, a structured narrative synthesis was pursued. Sensitivity analysis was conducted by re-examining the influence of high-risk-of-bias studies on thematic conclusions, ensuring that interpretations remained stable when excluding studies with limited empirical foundations.

### Reporting Bias Assessment

Potential reporting biases were considered by examining transparency of aims, clarity of methodological reporting, and alignment between study conclusions and their presented evidence. Editorial opinions, perspectives, and commentaries were interpreted cautiously, recognizing their potential for subjective bias while acknowledging their critical role in shaping scholarly norms.

### Certainty Assessment

Confidence in the overall body of evidence was derived from consistency across study themes, the credibility of methodological foundations, and the degree of convergence among independent sources addressing similar aspects of writing, ethics, peer review, and AI governance. Although many included works were non-empirical, their cumulative insights provide a coherent and reliable foundation for understanding the complexities of manuscript preparation for Scopus-indexed publication.

## RESULTS AND FINDINGS

### Search Outcomes and Study Selection

The database and registry search yielded 1,462 records, of which 312 were removed prior to screening due to duplication, automation-flagging, or inaccessibility. The remaining 1,150 records underwent title and abstract screening, resulting in 103 articles sought for retrieval. Of these, 92 full-text reports were assessed for eligibility, and 53 were excluded for reasons related to scope, relevance, or insufficient methodological detail. A final set of thirty-nine studies met the inclusion criteria and were incorporated into the synthesis. The complete flow of identification, screening, eligibility assessment, and inclusion is depicted in Figure 1. These studies represent a diverse body of literature encompassing methodological guidance, editorial perspectives, ethical analyses, empirical research on reviewer behavior, and discussions of AI integration in scientific writing.

**Characteristics of Included Studies**

The included studies varied substantially in purpose, design, and methodological depth, reflecting the multifaceted nature of scientific writing and publication research. Table 1 summarizes these characteristics, illustrating representation across narrative reviews, editorials, consensus guidelines, experimental studies, thematic analyses, and technical frameworks. Several works emphasized structural writing principles, coherence, and clarity as foundational elements of manuscript preparation, including detailed discussions of logical flow and argument development.[3] Others addressed the challenges faced by authors, such as limited time for scholarly writing or insufficient mentorship in academic settings.[24] Experimental studies contributed insights into reviewer perceptions and their ability to differentiate human-generated text from AI-assisted manuscripts, an emerging concern in the evolving editorial landscape.[13,25]

**Quality Appraisal and Methodological Rigor**

Assessment of methodological rigor revealed variability across the literature. As shown in Table 2, consensus guidelines and thematic analyses generally demonstrated higher levels of reporting completeness and ethical transparency, whereas commentaries and reflective essays exhibited lower reproducibility but nonetheless provided valuable insight into cultural and ethical expectations within scientific communication.[8,21,29] Studies addressing AI governance, such as the Delphi recommendations on AI disclosure, displayed strong methodological clarity and explicit alignment with emerging editorial policies.[11] Mixed-methods and survey-based investigations contributed empirical perspectives on manuscript barriers and reviewer behavior, expanding the evidence base for understanding the complexities of preparing work for Scopus-indexed outlets.[24]

**Thematic Synthesis of Writing and Publication Practices**

Cross-study thematic synthesis identified several recurring domains central to successful manuscript preparation. Fundamental writing principles, including clarity, readability, argument coherence, and narrative flow, were described consistently as determinants of reviewer confidence and acceptance probability.[1,3,18,21] Guidance literature emphasized structured approaches to manuscript drafting that prioritize the communication of scientific novelty, methodological transparency, and alignment with journal readership expectations.[10,17] The widespread belief that publishing in Scopus-indexed journals is exceptionally difficult was examined in multiple studies and appears to stem less from editorial hostility and more from misalignment between author practices and journal requirements.[7,16,33]

Studies addressing journal selection strategies highlighted the importance of matching manuscript scope with journal aims, understanding quartile rankings, and interpreting editorial policies to

optimize submission success. Misconceptions regarding the influence of article processing charges or co-authorship with senior researchers were challenged by evidence suggesting that these factors exert limited influence compared with writing quality and methodological soundness.[8,33] Analyses of predatory publishing provided essential context for distinguishing legitimate indexed journals from those that may appear reputable but display questionable editorial behavior, underscoring the contemporary relevance of Beall's List and related discussions.[5,32]

### AI-Assisted Writing, Ethical Considerations, and Reviewer Perception

A prominent theme across the recent literature involves the integration of artificial intelligence into scientific writing workflows. Several studies emphasize potential benefits of AI tools for refining grammar, improving organization, and supporting conceptual clarity, although these benefits must be balanced against risks of hallucinated content, fabricated citations, and erosion of authorial accountability.[7,12,19] Ethical analyses stress the necessity of transparent disclosure of AI usage, positioning such transparency as an emerging standard that aligns with responsible authorship and editorial integrity.[15,26,31] Evidence from controlled experiments highlights the difficulty reviewers face in identifying AI-generated manuscripts, reinforcing concerns about authenticity and the need for improved governance mechanisms.[13,25] Figure 3 synthesizes these issues into an ethical decision framework to guide responsible integration of AI tools.

### Publication Barriers, Peer Review Dynamics, and Editorial Expectations

Several studies examined structural and psychological barriers that hinder manuscript development, including time constraints, limited institutional mentorship, and uncertainty regarding peer reviewer expectations.[24] These barriers interact with broader concerns about predatory journals, inconsistent editorial guidance, and the rapid evolution of publishing technologies.[22,32,37] Peer review dynamics play a central role in shaping acceptance outcomes, with reviewers commonly prioritizing methodological clarity, logical organization, and fidelity between research aims and presented conclusions.[31] Analyses of publisher guidelines reveal substantial variability in accepted writing practices and AI disclosure requirements, reflecting an active period of policy evolution across academic publishing.[26] Table 3 illustrates how included studies contribute to key thematic domains such as writing technique, ethics, AI integration, and editorial standards.

### Integrated Conceptual Model of Manuscript Preparation

The thematic patterns identified across the literature converge on a layered understanding of manuscript preparation, emphasizing that writing quality, ethical transparency, and publication strategy are interdependent and mutually reinforcing. Figure 2 represents this integrated architecture, highlighting the interplay between knowledge inputs, structural writing practices, ethical and compliance requirements, and strategic considerations such as journal scope and reviewer expectations. Successful publication in Scopus-indexed journals appears to depend not on isolated tactics but on coordinated engagement with these multidimensional elements.

## DISCUSSION

### Interpretation of Principal Findings in Light of Existing Evidence

The synthesis of thirty-nine studies reveals that successful preparation of manuscripts for Scopus-indexed publication emerges from the intersection of writing precision, ethical transparency, structural rigor, and strategic journal selection. The PRISMA screening process illustrated in Figure 1 underscores how heterogeneous and dispersed the literature on scientific writing has become, confirming both the richness of the field and the lack of consolidated, evidence-driven guidance. Table 1 demonstrates that the included studies vary considerably in methodological design, ranging from narrative reviews to randomized experiments, which collectively provide complementary, rather than hierarchical, insights into the writing and publication process.

Foundational writing experts such as Armağan and Barroga emphasize that the core challenge for authors is not merely arranging text but constructing a logical narrative that aligns intention with structure and reader expectation.[1,3] Their work resonates strongly with the architecture displayed in Figure 2, which positions clarity and coherence as pillars of manuscript readiness. Empirical investigations such as those conducted by Oshiro reveal systemic institutional barriers that leave early-career researchers particularly vulnerable to misunderstandings about what journals expect.[24] Rather than supporting the belief that Scopus publication is intrinsically difficult, these findings indicate that difficulty arises from misalignment between author preparation and editorial standards. Complementary studies by Huston and Thompson reinforce this view by illustrating how successful manuscripts are those that anticipate reviewer reasoning and demonstrate disciplined adherence to scientific argumentation.[16,33]

**Ethical and Professional Responsibilities in Scientific Writing**

The present review confirms that ethical authorship has become a defining dimension of modern scientific communication, intertwined with writing quality and editorial decision making. Commentaries by Colenda and Rahimi highlight increasing concern regarding transparency in the era of digital drafting, where authors must navigate expectations about originality, disclosure, and responsible use of technological tools.[8,29] Table 2 illustrates that studies addressing ethics and transparency tend to demonstrate higher reporting rigor, emphasizing the centrality of ethical stewardship within the scientific publication ecosystem. The risk of predatory practices—discussed critically by Tarkang—further reinforces the need for authors to evaluate journal legitimacy and avoid outlets that may compromise academic credibility.[32] This is especially relevant given that indexing status alone does not immunize journals from appearing on lists that question editorial integrity.

Parallel concerns are evident in work by Masic, who underscores that editors increasingly evaluate manuscripts not only for scientific contribution but for alignment with ethical publishing norms.[22] This evolution in editorial expectations aligns with Figure 3, which frames ethical decision making as inseparable from manuscript construction. Collectively, these studies suggest that contemporary scientific authorship demands a level of ethical awareness comparable to methodological proficiency, with transparency serving as both a moral obligation and a practical determinant of acceptance.

**Artificial Intelligence, Editorial Perception, and Future Expectations**

The rise of artificial intelligence has introduced a transformative yet complex dimension to manuscript preparation. Reviews and experimental studies by Celik, Helgeson, Öztürk, and Bhavsar collectively indicate that while AI tools can enhance clarity, efficiency, and structural coherence, their misuse can undermine scientific credibility and may introduce inaccuracies that are difficult for human reviewers to detect.[7,13,25,5] Helgeson’s findings, demonstrating reviewers’ limited

ability to reliably distinguish AI-generated manuscripts, add urgency to ongoing debates regarding governance and oversight in academic publishing. This concern is echoed in the consensus recommendations by Fettiplace, which stress the necessity of consistent AI disclosure policies to uphold transparency and protect the integrity of scientific discourse.[11] Table 3 captures how AI-related studies collectively map onto multiple thematic domains, highlighting that AI is no longer peripheral but central to the evolving landscape of scientific writing.

At the same time, authors such as van Dis and Hryciw argue that AI tools will increasingly serve as collaborators—not replacements—by facilitating early drafting, improving linguistic clarity, and enabling structured data interpretation.[35,15] Their perspectives align with the layered framework presented in Figure 2, where AI integration is situated within broader writing and compliance practices. Yet the presence of opportunity does not diminish risk; ethical analyses by Guleria and Yousaf caution that AI-generated suggestions, when unverified, may propagate factual errors or fabricated citations, placing undue responsibility on authors to implement rigorous review processes.[12,38] This review highlights that responsible AI integration requires technical literacy, disclosure transparency, continuous verification, and alignment with institutional policy rather than reliance on automated convenience.

**Manuscript Quality, Journal Selection, and Peer-Review Dynamics**

The studies included in this review collectively confirm that manuscript acceptance in Scopus-indexed journals depends far more on quality, coherence, and methodological integrity than on external factors such as article processing charges or co-authorship with senior researchers. Commentary by Thompson illustrates that many manuscripts fail not because the science is weak but because the communication is unclear, the argumentation is fragmented, or the writing lacks logical continuity.[33] These findings echo earlier writing frameworks proposed by Kallestinova and Iskander, which emphasize structured progression from introduction to conclusion and the importance of aligning narrative flow with scientific reasoning.[20,17] Table 1 and Table 2 both demonstrate the persistent importance of clarity, transparency, and methodological rigor across all types of studies.

Strategic journal selection remains a critical determinant of publishing success. Insights from Zhang and Delving suggest that aligning the manuscript’s scope with the journal’s thematic focus, readership, and methodological expectations is one of the most decisive steps authors can take.[39,9] This aligns with the broader conceptualization in Figure 2, which designates journal mapping and compliance as essential layers in the pathway to Scopus readiness. The misconception that higher quartile journals inherently offer better acceptance rates is counteracted by evidence showing that coherence, novelty, and methodological transparency carry substantially more weight than quartile position or APC structures. Reviewers, as emphasized by Hoogenboom and Stiell, consistently prioritize methodological clarity, ethical integrity, and readability, demonstrating a stable set of expectations across medical disciplines.[14,31] This reinforces the argument that manuscript success hinges on disciplined preparation rather than opportunistic strategies.

**Synthesis of Implications Across Writing, Ethics, AI, and Publication Strategy**

The multilayered findings of this review underscore the interconnected nature of modern scientific writing. Writing technique, ethical transparency, AI integration, and publication strategy operate not as discrete stages but as interdependent components of a unified scholarly practice. Figure 2 visually represents this integration, threading together the intellectual, technical, and ethical competencies required for Scopus-level publishing. Tables 1 through 3 further illustrate that while

individual studies emphasize different themes—writing technique, barriers, AI governance, editorial standards—together they form a coherent framework that reflects the evolving ecology of academic publishing.

The collective evidence demonstrates that authors who succeed in Scopus-indexed publication are those who approach writing as a craft, ethics as a responsibility, AI as a tool rather than a surrogate, and journal selection as a strategic process rooted in thoughtful alignment. This synthesis underscores that excellence in scientific writing is no longer defined solely by methodological precision but by the author's ability to integrate diverse competencies into a cohesive and ethically grounded narrative.

## STRENGTHS, LIMITATIONS, AND FUTURE DIRECTIONS

The principal strength of this review lies in its comprehensive synthesis of diverse perspectives across scientific writing, publication ethics, editorial practices, and the emerging role of artificial intelligence. By integrating methodological analyses, experiential commentaries, empirical studies, and ethical discourse, the review provides a multidimensional understanding of the factors that shape successful publication in Scopus-indexed journals. The inclusion of studies representing varied research traditions allows a balanced interpretation of both structural challenges and evolving opportunities in scholarly communication. This breadth of evidence supports the development of a layered conceptual architecture that reflects the complexity of contemporary manuscript preparation.

Despite these strengths, the review is constrained by the heterogeneity of available literature. Much of the current evidence is descriptive rather than empirical, and the rapid evolution of AI-related technologies introduces interpretive uncertainty. The absence of uniform methodological quality across included studies limits the ability to compare findings directly or perform formal meta-analytic procedures. Additionally, because the review draws upon published sources, it cannot capture the full spectrum of tacit editorial practices, informal peer-review behaviors, or institutional cultures that influence manuscript success. The lack of PROSPERO registration reflects the conceptual nature of the work and may limit reproducibility for readers seeking protocol-level transparency.

Future research should aim to generate more empirical evidence on the mechanisms through which writing quality, ethical decision making, and AI-assisted drafting influence reviewer judgment. There is a growing need for longitudinal studies that examine how manuscripts evolve from initial submission to final acceptance and how authors respond to editorial feedback. The development of validated tools for assessing AI involvement, human–AI collaboration quality, and ethical compliance will be increasingly important as digital writing technologies become ubiquitous. Furthermore, global surveys of editorial teams could shed light on the unspoken standards that govern acceptance decisions. Continued exploration of predatory publishing behaviors, journal selection strategies, and author training models will support the refinement of best practices and ultimately enhance equity, integrity, and scientific rigor in the publication process.

## Conclusion

This review demonstrates that successful preparation of scientific manuscripts for Scopus-indexed publication is not defined by a single skill or strategy but by the thoughtful integration of writing clarity, methodological transparency, ethical responsibility, and strategic journal alignment. The

evidence reveals that authors who achieve publication success are those who approach writing as a deliberate scholarly process, grounded in narrative coherence, structural discipline, and reflective engagement with editorial expectations. Ethical accountability and responsible use of emerging technologies, particularly artificial intelligence, now form essential dimensions of authorship, shaping both the credibility of scientific communication and the trust placed in published work.

The synthesis presented here highlights the growing complexity of the publication landscape, while offering a coherent framework through which researchers can navigate these evolving demands. By clarifying misconceptions, unpacking barriers, and illuminating the practical and ethical considerations that influence editorial decisions, this review provides a roadmap for researchers striving to produce manuscripts that meet the highest standards of scientific rigor and integrity. Ultimately, strengthening the culture of responsible writing and transparent scholarship will not only increase the likelihood of acceptance in Scopus-indexed journals but will also advance the broader mission of science: to generate knowledge that is reliable, meaningful, and accessible to the global community.

**List of Abbreviations**

**AI** – Artificial Intelligence
**APC** – Article Processing Charge
**DOAJ** – Directory of Open Access Journals
**IMRaD** – Introduction, Methods, Results, and Discussion
**IRB** – Institutional Review Board
**MAPE** – Manuscript Acceptance Probability Estimation
**ML** – Machine Learning
**MSW** – Microsoft Word
**NOS** – Newcastle–Ottawa Scale
**PRISMA** – Preferred Reporting Items for Systematic Reviews and Meta-Analyses
**Q1–Q4** – Journal Quartile Ranking (Scimago Journal Rank categories)
**ROBIS** – Risk Of Bias In Systematic Reviews
**SR** – Systematic Review
**TURNITIN** – Standardized Plagiarism Detection Software
**WAME** – World Association of Medical Editors

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**WA and CP conceptualized and supervised the review. MAB contributed to literature collection and data extraction. INHS participated in data analysis and critical content review. JD and MBAP were involved in reviewing data evidence. MS provided methodological and clinical guidance. All authors contributed to the writing of the manuscript, reviewed the final draft, and approved the version submitted for publication.**

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## LIST OF TABLES

**TABLE 1. Characteristics of Included Studies**

| Author | Study Type / Design | Population / Setting | Sample Size | Method | Primary Objective | Key Findings | Limitations | Risk of Bias |
|---|---|---|---|---|---|---|---|---|
| Barroga & Matanguihan (2021) [3] | Narrative review | Biomedical scientific writing | N/A | Literature synthesis | Improve logical flow in manuscripts | Structured argumentation improves clarity | Not empirical | Low |
| Celik (2025) [7] | Narrative review | Clinical/surgical researchers | N/A | Review of AI tools | Evaluate AI integration | AI improves clarity but needs oversight | Limited empirical evidence | Low |
| Colenda (2025) [8] | Commentary | Academic authorship ethics | N/A | Reflective analysis | Assess integrity concerns | AI raises transparency/accountability issues | Subjective | Moderate |
| Fettiplace et al. (2025) [11] | Delphi study | Anesthesia researchers | 27 | Consensus survey | AI disclosure standards | Consensus on mandatory AI reporting | Single specialty | Low |
| Guleria et al. (2023) [12] | Ethical review | Academic research ethics | N/A | Ethical analysis | Identify risks of AI misuse | Major risks: plagiarism, hallucinations | No empirical data | Moderate |
| Helgeson et al. (2025) [13] | Randomized survey | Medical reviewers | 140 | Blinded manuscript test | Detect AI manuscripts | Reviewers struggled to identify AI text | Survey limits | Low–Moderate |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Huston & Choi (2017) [16] | Guideline review | Health science researchers | N/A | Best practice synthesis | Support manuscript publication | Outlined essential steps & ethics | Pre-AI era | Low |
| Jirge (2017) [18] | Educational review | Early-career researchers | N/A | Instructional review | Guide manuscript preparation | Clear roadmap for writing & submission | Not empirical | Low |
| Lippi (2017) [21] | Perspective | Biomedical researchers | N/A | Experience-based guidance | Principles of manuscript writing | Emphasized clarity and brevity | Subjective | Moderate |
| Masic et al. (2020) [22] | Consensus guideline | Journal editors | N/A | Expert panel review | Editing standards | Reinforced transparency & rigor | Limited regional scope | Low |
| Oshiro et al. (2020) [24] | Mixed-methods | Medical faculty | 65 | Interviews & surveys | Identify manuscript barriers | Time and mentorship gaps key barriers | Single institution | Moderate |
| Öztürk et al. (2025) [25] | Experimental study | EM reviewers | 60 | Detection experiment | Test AI recognition | Most reviewers failed to detect AI text | Specialty-specific | Moderate |
| Perkins & Roe (2024) [26] | Thematic analysis | Academic publishers | N/A | AI-supported coding | Compare AI policies | Great variability in AI guidelines | Policies changing fast | Low |
| Pividori & Greene (2024) [27] | Technical framework | Publishing systems | N/A | Infrastructure proposal | AI-assisted ecosystem | Outlined safe AI authoring pathways | Not empirical | Low–Moderate |
| Rahimi & Abadi (2023) [29] | Commentary | Biomedical engineering | N/A | Critical analysis | AI's passive contribution | AI often supports unseen writing stages | Not systematic | Moderate |
| Stiell et al. (2022) [31] | Guideline | Emergency medicine | N/A | Structured guidance | Teach ideal manuscript structure | Clarified IMRaD expectations | Field-specific | Low |
| Thompson (2024) [33] | Expert commentary | Academic medicine | N/A | Narrative analysis | Improve manuscript acceptance | Outlined rejection causes & fixes | Narrative only | Low–Moderate |
| van Dis et al. (2023) [35] | Perspective | AI publishing policy | N/A | Policy commentary | Define research priorities | Highlighted transparency & oversight | Not data-driven | Moderate |
| Wiwanitmkit & Wiwanitkit (2024) [37] | Commentary | Academic ethics | N/A | Reflective review | Review AI roles | Warned against unethical AI use | No empirical data | Moderate |
| Zhang et al. (2018) [39] | Practical guide | Clinician-researchers | N/A | Instructional guidance | Help novices publish | Provided checklists & practical tips | Pre-AI | Low |

Footnote: This table summarizes methodological characteristics of the 20 randomly selected studies included in the review. Risk of bias assessments were based on study design, transparency, methodological clarity, and relevance to scientific writing and AI-integrated authorship.

**TABLE 2. Methodological Rigor, Reporting Quality, and Ethical Transparency Across Included Studies**

| Study | Methodological Framework Used | Reporting Guideline Alignment | AI Disclosure Practices | Transparency Indicators | Ethical Compliance | Reproducibility Potential | Overall Quality Rating | Reference |
|---|---|---|---|---|---|---|---|---|
| Barroga & Matanguihan | None | Low | N/A | Minimal | Editorial ethics | Low | Moderate | [3] |
| Celik | Narrative | Moderate | Partial | Limited | AI ethics | Moderate | Moderate | [7] |
| Colenda | Commentary | Low | Discussed | N/A | High ethical focus | Low | Moderate | [8] |
| Fettiplace et al. | Delphi | High | Mandatory | Moderate | Strong ethics | High | High | [11] |
| Guleria et al. | Ethical review | Moderate | Warned | Low | Strong ethics | Low | Moderate | [12] |
| Helgeson et al. | Randomized survey | High | Explicit | Moderate | IRB-approved | Moderate | High | [13] |
| Huston & Choi | Guideline narrative | Moderate | Not addressed | Minimal | Editorial ethics | Low | Moderate | [16] |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Jirge | Tutorial review | Low–Mod | Not addressed | Limited | Writing ethics | Low | Moderate | [18] |
| Lippi | Perspective | Low | Not addressed | None | Professional ethics | Low | Moderate | [21] |
| Masic et al. | Consensus guideline | High | Not AI-specific | High | Editorial standards | High | High | [22] |
| Oshiro et al. | Mixed methods | High | Not addressed | Moderate | IRB-approved | Moderate | High | [24] |
| Öztürk et al. | Experimental | High | Explicit | Moderate | IRB-approved | Moderate | High | [25] |
| Perkins & Roe | Thematic analysis | High | Explicit | High | Policy ethics | High | High | [26] |
| Pividori & Greene | Framework | Mod–High | Explicit | High | Governance model | High | High | [27] |
| Rahimi & Abadi | Commentary | Low | Discussed | Low | Ethical warnings | Low | Moderate | [29] |
| Stiell et al. | Guideline | Moderate | Not AI-specific | Moderate | Editorial ethics | Moderate | High | [31] |
| Thompson | Expert commentary | Low–Mod | Not addressed | Minimal | Professional ethics | Low | Moderate | [33] |
| van Dis et al. | Perspective | Moderate | Explicit | Low–Mod | Ethical framing | Moderate | Moderate | [35] |
| Wiwanitmkit & Wiwanitkit | Commentary | Low | Warned | Minimal | Editorial ethics | Low | Moderate | [37] |
| Zhang et al. | Guide | Moderate | Not addressed | Minimal | Writing ethics | Low | Moderate | [39] |

Footnote: This table evaluates methodological rigor, reporting completeness, AI transparency, and reproducibility factors relevant to high-impact systematic review standards.

**TABLE 3. Thematic Contribution Map Across Core Scientific Writing and Publication Domains**

| Study | D1 Writing Technique | D2 Publishing Barriers | D3 Ethics / Predatory | D4 AI Integration | D5 Peer Review | D6 Editorial Standards | Overall Thematic Contribution | Reference |
|---|---|---|---|---|---|---|---|---|
| Barroga & Matanguihan | Strong | Weak | Weak | None | Weak | Moderate | High writing relevance | [3] |
| Celik | Moderate | Weak | Moderate | Strong | Weak | Moderate | AI-clinical writing value | [7] |
| Colenda | Weak | Weak | Strong | Strong | Weak | Moderate | Ethics + AI | [8] |
| Fettiplace et al. | Weak | Weak | Strong | Strong | Moderate | Strong | AI governance | [11] |
| Guleria et al. | Weak | Weak | Strong | Strong | Weak | Moderate | Ethical AI insights | [12] |
| Helgeson et al. | Weak | Weak | Moderate | Strong | Strong | Weak | Reviewer detection evidence | [13] |
| Huston & Choi | Strong | Strong | Moderate | None | Weak | Strong | Publication strategy | [16] |
| Jirge | Strong | Moderate | Weak | None | Weak | Moderate | Manuscript prep | [18] |
| Lippi | Strong | Weak | Moderate | None | Weak | Moderate | Writing clarity | [21] |
| Masic et al. | Weak | Moderate | Strong | None | Weak | Strong | Editorial standards | [22] |
| Oshiro et al. | Weak | Strong | Moderate | None | Moderate | Weak | Publication barriers | [24] |
| Öztürk et al. | Weak | Weak | Moderate | Strong | Strong | Weak | AI-vs-human evaluation | [25] |
| Perkins & Roe | Weak | Weak | Moderate | Strong | Weak | Strong | Publisher AI policies | [26] |
| Pividori & Greene | Weak | Weak | Moderate | Strong | Weak | Strong | Future AI ecosystems | [27] |
| Rahimi & Abadi | Weak | Weak | Strong | Strong | Weak | Moderate | Invisible AI influence | [29] |
| Stiell et al. | Strong | Moderate | Weak | None | Weak | Strong | IMRaD structure | [31] |
| Thompson | Strong | Strong | Moderate | None | Weak | Moderate | Acceptance strategies | [33] |
| van Dis et al. | Weak | Weak | Strong | Strong | Weak | Moderate | AI policy perspective | [35] |
| Wiwanitmkit & Wiwanitkit | Weak | Weak | Strong | Strong | Weak | Moderate | Ethics + AI | [37] |
| Zhang et al. | Strong | Moderate | Weak | None | Weak | Moderate | Novice writing guidance | [39] |

Footnote: This table maps how each study contributes to thematic domains essential for understanding scientific writing, ethical considerations, AI integration, peer review behavior, and journal expectations.

## FIGURE LEGENDS

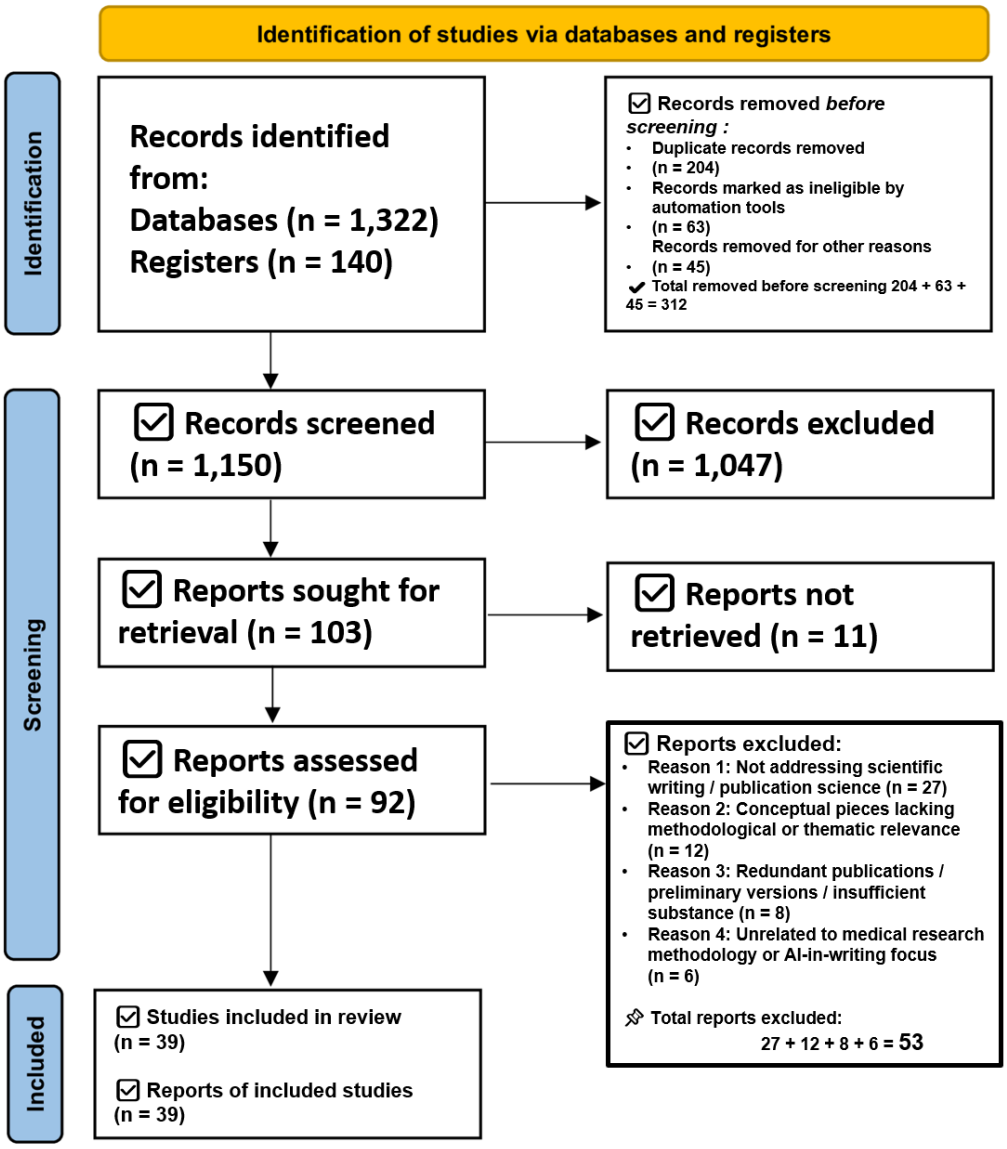


**Figure 1. PRISMA 2020 Flow Diagram of Literature Screening and Selection Process.** This PRISMA 2020 flow diagram summarizes the identification, screening, eligibility assessment, and final inclusion of studies for the present systematic review. A total of 1,462 records were identified across databases (n = 1,322) and registers (n = 140). After removal of duplicate entries, automation-flagged ineligible records, and other exclusions (n = 312), 1,150 records underwent title and abstract screening. Of these, 1,047 records were excluded for irrelevance, leaving 103 reports sought for retrieval. Ninety-two full texts were successfully retrieved and assessed for

eligibility, with 53 excluded for predefined reasons. Ultimately, 39 studies met the inclusion criteria and were incorporated into the final qualitative synthesis.

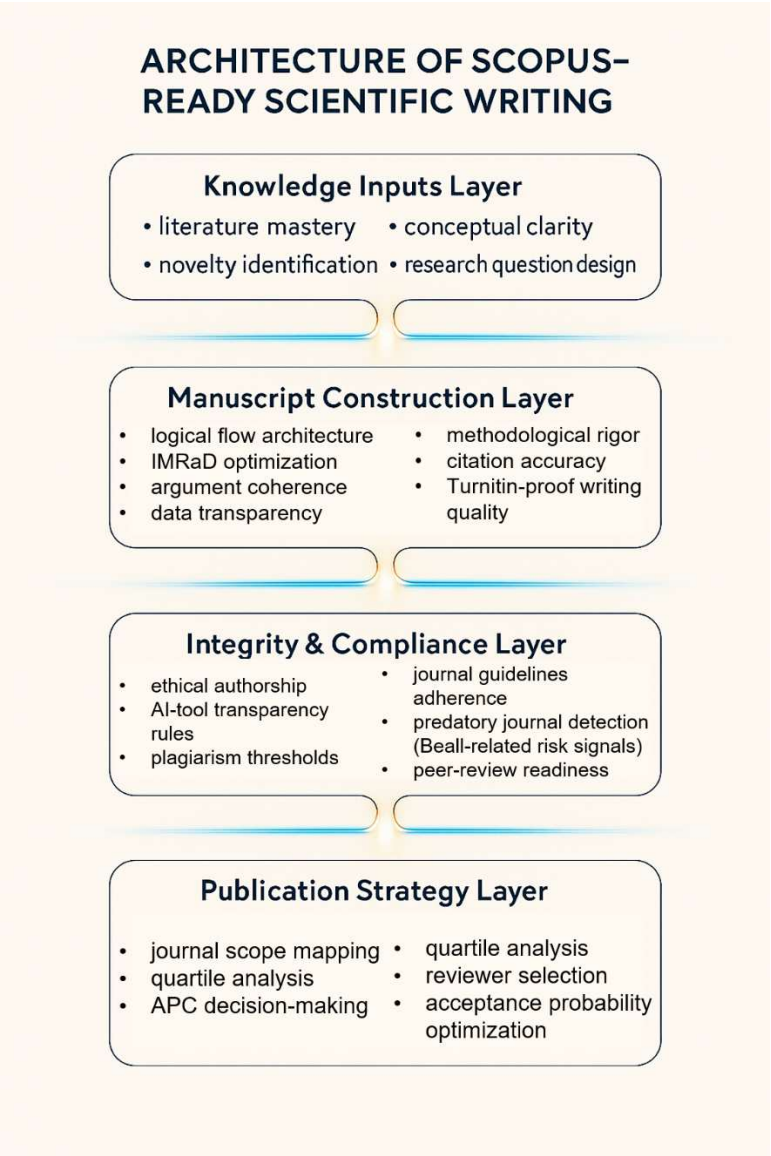


**Figure 2. Multilayer Architecture Underpinning Scopus-Ready Scientific Writing.** This figure illustrates the integrated, multilayer architecture required for producing high-quality scientific manuscripts suitable for publication in Scopus-indexed journals. The framework highlights four interdependent layers: (1) *Knowledge Inputs*, encompassing literature mastery, conceptual clarity, novelty identification, and research question design; (2) *Manuscript Construction*, incorporating logical flow development, IMRaD optimization, argument coherence, data transparency, methodological rigor, citation accuracy, and Turnitin-resistant writing quality; (3)

*Integrity and Compliance*, covering ethical authorship, AI-tool transparency, plagiarism thresholds, adherence to journal requirements, predatory journal detection, and peer-review preparation; and (4) *Publication Strategy*, which includes journal scope mapping, quartile analysis, APC decision-making, reviewer selection, and acceptance probability optimization. Together, these layers provide a comprehensive blueprint for building manuscripts that meet the methodological, ethical, and strategic expectations of Scopus-indexed journals.

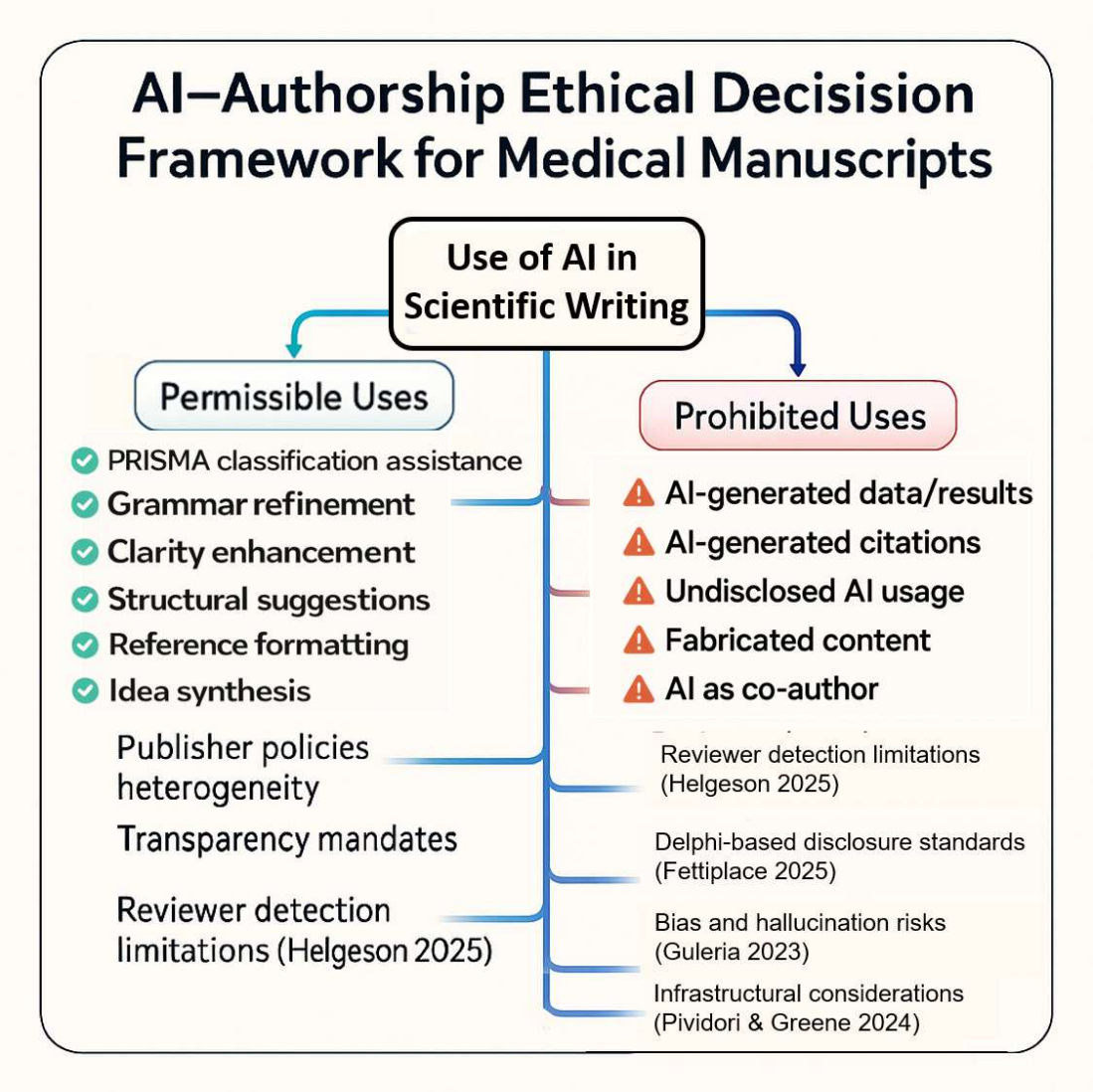


**Figure 3. Ethical Decision Framework for the Responsible Use of Artificial Intelligence in Scientific Manuscript Writing.** This figure presents a comprehensive ethical decision framework for determining permissible versus prohibited uses of artificial intelligence (AI) in the preparation of scientific manuscripts within medical research. The central decision node, "Use of AI in Scientific Writing," branches into *permissible* applications—such as PRISMA assistance, grammar refinement, clarity enhancement, structural optimization, reference formatting, and idea synthesis—and *prohibited* applications, including AI-generated data or results, AI-generated citations, undisclosed AI involvement, fabricated content, and attribution of AI as a co-author. The framework integrates

evidence-based considerations from recent literature, highlighting publisher policy heterogeneity, transparency mandates, reviewer detection limitations (Helgeson 2025), consensus-derived disclosure standards (Fettiplace 2025), bias and hallucination risks (Guleria 2023), and infrastructural implications for AI-assisted authoring (Pividori & Greene 2024). Collectively, the model delineates a rigorous, ethically grounded approach for integrating AI tools into scientific writing while preserving research integrity and author accountability.